

Figure 18.4 Types of chromosomes based on position of centromere

18.5.2 Types of Chromosomes based on the position of Centromere

Based on the position of centromere, the chromosomes are classified as **Telocentric**, **Acrocentric, Submetacentric** and **Metacentric**

1. **Telocentric**– The centromere is found on the proximal end. They are rod shaped chromosomes.
2. **Acrocentric** – The centromere is found at the one end with a short arm and a long arm. They are also rod-shaped chromosomes.
3. **Submetacentric** – The centromere is found near the centre of the chromosome. Thus forming two unequal arms. They are J shaped or L shaped chromosomes.
4. **Metacentric** – The centromere occurs in the centre of the chromosome and form two equal arms. They are V shaped chromosomes

18.5.3 Types of Chromosomes based on function

The eukaryotic chromosomes are classified into **autosomes** and **allosomes.**

Autosomes contain genes that determine the somatic (body) characters**.** Male and female have equal number of autosomes**.**

Allosomes are chromosomes which are responsible for determining the sex of an individual. They are also called as **sex chromosomes** or **hetero-chromosomes**. There are two types of sex chromosomes, X and Y- chromosomes. Human males have one X chromosome and one Y chromosome and human females have two X chromosomes.

18.5.4 Karyotype

The number of chromosomes in any living organism (animal or plant) is constant. In human, each cell normally contains **23 pairs** of chromosomes. Out of which 22 pairs are autosomes and the 23rd pair is the allosome or sex chromosome.

In the body cells of sexually reproducing organisms, the chromosomes generally occur in pairs. This condition is called **diploid (2n)**. The gametes produced by the organisms contain a single set of chromosomes. Hence, the gametes are said to be **haploid (n).**

Karyotype is the **number, size and shape of chromosomes** in the cell nucleus of an organism. **Idiogram** is the diagrammatic representation of karyotype of a species. It consists of all the metaphasic chromosomes arranged in homologous pairs according to decreasing length, thickness, position of centromere, shape etc., with the sex chromosomes placed at the end.

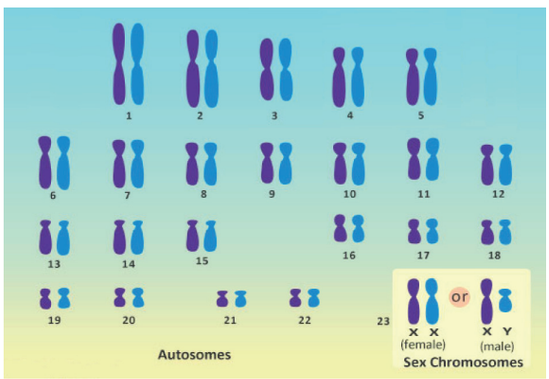


Figure 18.5 Normal human karyotype

18.6 Structure of DNA

DNA is the hereditary material as it contains the genetic information. It is the most important constituent of a chromosome. The most widely

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